Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Arrival · Aspire

1. Enzyme and substrate

Why does an iodine colour change provide indirect evidence of amylase activity?

Iodine detects starch in a sample; it does not show the enzyme molecules working.

2. Read the colour

At 60 s starch is detected; at 90 s it is not. Is an exact endpoint of 75 s justified?

No observation was made at 75 s. Distinguish an observed endpoint from its possible timing window.

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3. Change and measure

Distinguish controlled temperature from a comparison with no active amylase.

A controlled variable stays matched. A comparison without active enzyme checks whether the enzyme is needed for the observed change.

4. Prepare the evidence

The 60 s sample is missed. The endpoint is seen at 90 s. Can that endpoint still have the same 30 s timing window?

The last sample that showed starch might now be at 30 s. Use only times that were seen.

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Exit Ticket · Aspire

Evidence

These are example results at pH 7: 30 s blue-black; 60 s blue-black; 90 s orange-brown. Blue-black means starch was detected. Orange-brown means no starch was detected in that sample. The planned interval is 30 s. The amounts at the start match. The temperature and the endpoint rule match too.

1. If the 60 s sample was missed, how would your endpoint statement and timing window change?

Use only 30 s positive and 90 s negative; do not invent the missing result.

2. Name one authorised way to improve this timing limit. Name one uncertainty that it does not remove.

Link the improvement to the gap between sample times. Then think about judging colour.

Under the heading Exit ticket: answer the short questions in your book.